HANDS-ON 5 · DAY 2 / MORNING · GROUP'S SHEET

Which reviews should the care team read first?

The case. A streaming service receives more reviews than its care team can read. The team wants the unhappy ones first: missing an unhappy review costs it four times as much as reading a happy one by mistake. Can a model trained on 2,000 labelled reviews pick the reviews to read? And does it do as well on 2,000 reviews from another source?

DATA CARD · IMDB-REVIEWS.CSV

What a row is. One review of one film, written by an IMDb user. 2,000 rows, 3 columns, no missing value: `Text`, the review; `URL`, the film's IMDb page; `unhappy`, the target. The median review has 177 words, the shortest 14, the longest 1,827.

The target. `unhappy`: 1 for a negative review (1,000 rows), 0 for a positive one (1,000), so always guessing one class is right half the time.

How the rows were chosen. A subset of the 50,000 labelled reviews of Maas et al., who kept only clear verdicts (4 or less out of 10 for a negative review, 7 or more for a positive one) and at most 30 reviews a film. Here 1,247 films, up to 7 reviews each: 1,225 reviews share their film with another.

Its history. The inherited copy repeated the opening sentence at the end of 1,988 texts, 168 of those copies followed by a cut HTML tag `<br`: both removed. 161 texts stated a rating ("8/10", "9 out of 10"), which gives the label away: stripped. The labels `POS` and `NEG` became `unhappy`; the column `Index`, not a row key, was dropped.

Source and licence. A. L. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng and C. Potts (2011), Learning word vectors for sentiment analysis, ACL-HLT 2011, 142 to 150. No licence is recorded for the texts: used in class, never republished. From panel 4: B. Pang and L. Lee (2004), polarity dataset v2.0, reviews from before 2002, already in lower case; 60 of them stated a rating, stripped.

file	reviews	unhappy	columns	used in
`imdb-reviews.csv`	2,000	1,000	`URL`, `Text`, `unhappy`	steps 1 to 3
`polarity-reviews.csv`	2,000	1,000	`filename`, `Text`, `unhappy`	panel 4

STEP 1 · PREPARE THE REVIEWS

1 ☐ **Upload the reviews**
In a new project: **+ Dataset** → **Upload your files**, `imdb-reviews.csv`.

Rows: _____ unhappy: _____

2 ☐ **Simplify the text**
On the dataset: **+ Recipe** → **Prepare**, a step **Simplify text** on `Text`, output column `TextSimple`, with **Normalize text**, **Stem words** and **Clear stop words**, language English; **Run**.

3 ☐ **Check what happens to "not"**
In the Prepare preview, find a review whose `Text` holds "not" and compare it with its `TextSimple`.

Is "not" still there? _____ If not, what did you change? _____

STEP 2 · TRAIN A MODEL ON THE WORDS

1 ☐ **Create the analysis**
Select the prepared dataset, then **Lab** → **AutoML prediction**: target `unhappy`, two classes.

2 ☐ **Read only the simplified text**
Design → **Features handling**: `Text` and `URL` rejected; `TextSimple` as text, **Text handling TF/IDF vectorization**, min. rows fraction 0.1%, max. rows fraction 80%, n-grams 1 to 2.

3 ☐ **Keep each film in one fold**
Design → **Train/Test set**: **K-fold cross-test**, 5 folds, grouped by `URL`. **Design** → **Algorithms**: **Logistic regression** only. **Train**.

Training time: _____

STEP 3 · READ IT AT THE CARE TEAM'S COSTS

1 ☐ **Read the score and its spread**
Results → **Performance**: the ROC AUC and its spread across the 5 folds.

AUC: _____ ± _____

2 ☐ **Set the threshold the costs give**
On the confusion matrix, move the threshold to 0.2; read the four cells, then the same at 0.5.

threshold	TP	FP	FN	TN	cost 4 × FN + FP
0.2					
0.5					

3 ☐ **Read the heaviest terms**
Results → **model** → **Regression coefficients**: the terms with the largest weights, each way.

Towards unhappy: _____ towards happy: _____

Are they opinions, or the names of films and actors?

4 · SCORE REVIEWS FROM ANOTHER COLLECTION

1 ☐ **Prepare the new reviews alike**
Upload `polarity-reviews.csv`; copy the Prepare recipe of step 1 onto it, so that it gets its own `TextSimple`; **Run**.

2 ☐ **Score them with your model**
In the analysis: **Deploy**. On the prepared polarity reviews: **+ Recipe** → **Score** with that model, then **Run**. Is the threshold applied 0.2? _____

3 ☐ **Count its errors**
On the scored dataset, a Prepare step **Formula** into `cell`: `if(prediction == 1, if(unhappy == 1, "TP", "FP"), if(unhappy == 1, "FN", "TN"))`; then count the values of `cell`.

reviews	TP	FP	FN	TN	cost 4 × FN + FP	cost per 100 reviews
IMDb (step 3.2, 0.2)						
Pang and Lee						

Does your model carry to the other collection? What differs between the two (data card)?

5 · DECIDE

Would you give your model to the care team, and at which threshold? What would you measure first on the team's own reviews before it goes live?